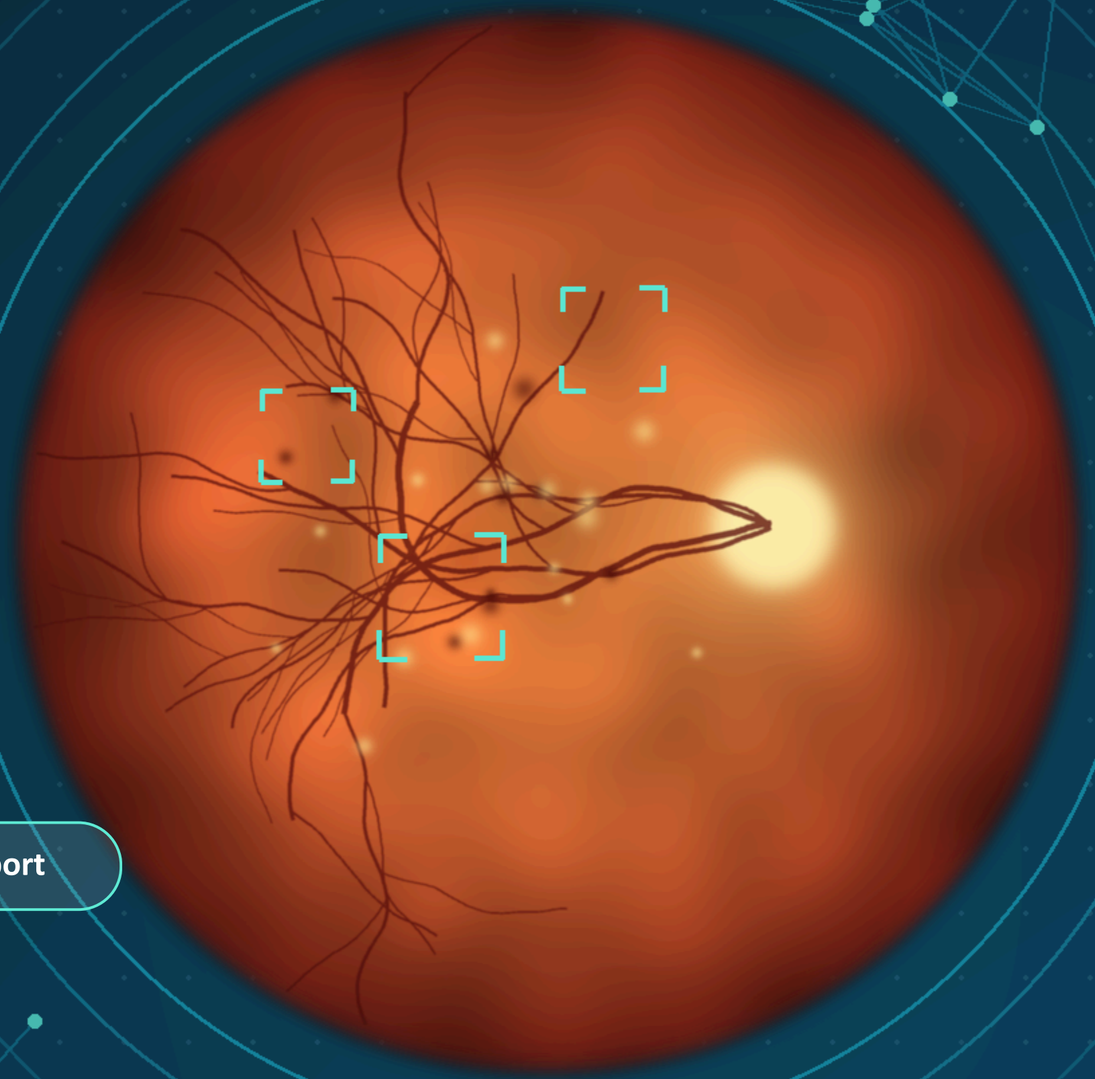
FIRST REVIEW · PROJECT PROGRESS

DR Screening Suite

Explainable diabetic retinopathy screening, combined with metabolic risk



Retina AI | Grad-CAM | Lesion maps | Risk report

Team: Srushti , Sanobar, Prajakta

Why this matters

A silent disease

Diabetic retinopathy damages the tiny blood vessels of the retina, often with no early symptoms.

Screening is slow

Eye specialists are few, and grading every retina photo by hand takes time.

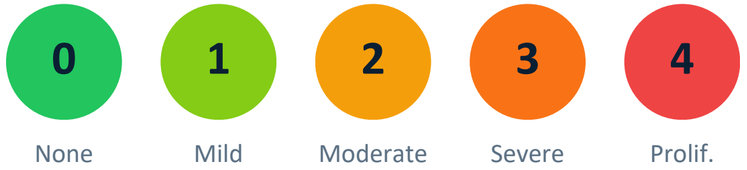
Early detection saves sight

When it is caught early, vision loss can be prevented or slowed down.

1 in 3

people with diabetes develop some degree of retinopathy

Five grades: 0 (healthy) to 4 (most severe)



THE SOLUTION

What our app does

Grades it

Reads a retina photo and predicts the DR grade, 0 to 4.

Explains it

A heatmap shows where the AI looked to decide.

Locates it

Marks lesions, the optic disc and the fovea.

Checks metabolic risk

Uses HbA1c, blood pressure and diabetes duration.

Reports it

One combined risk level and a PDF report.

Upload a retina photo → get an explained result in seconds

THE PIPELINE

How it works

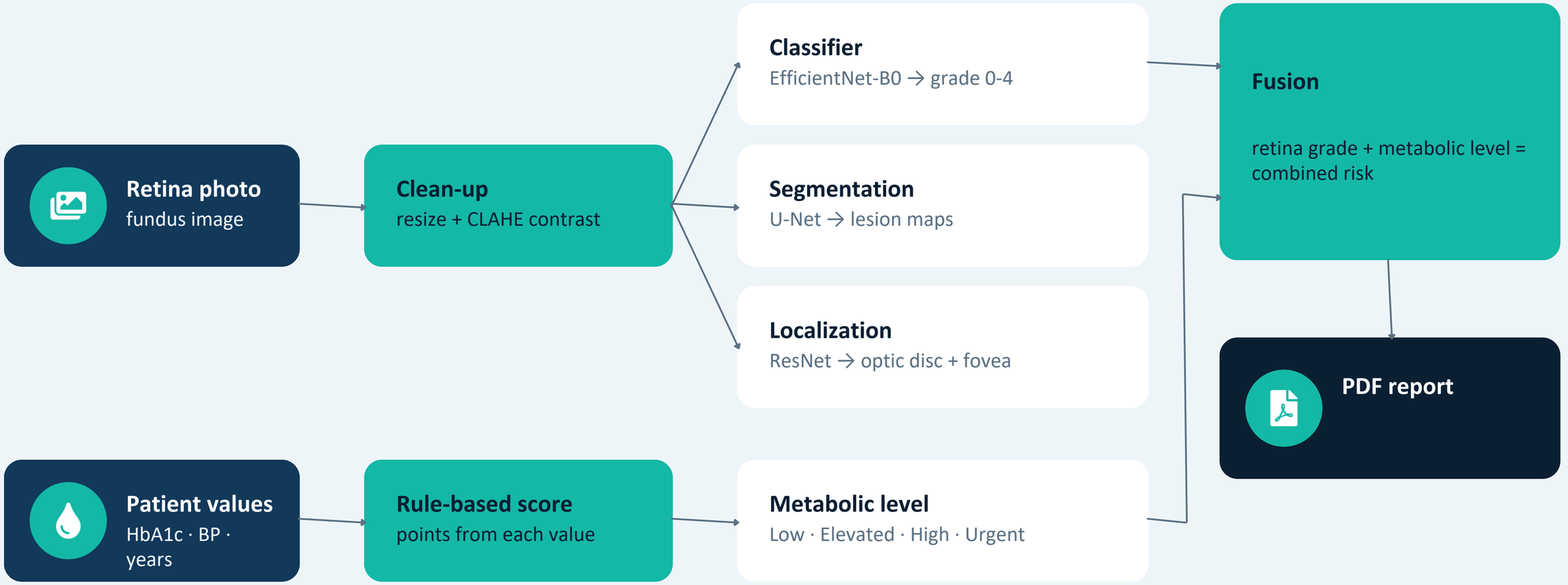
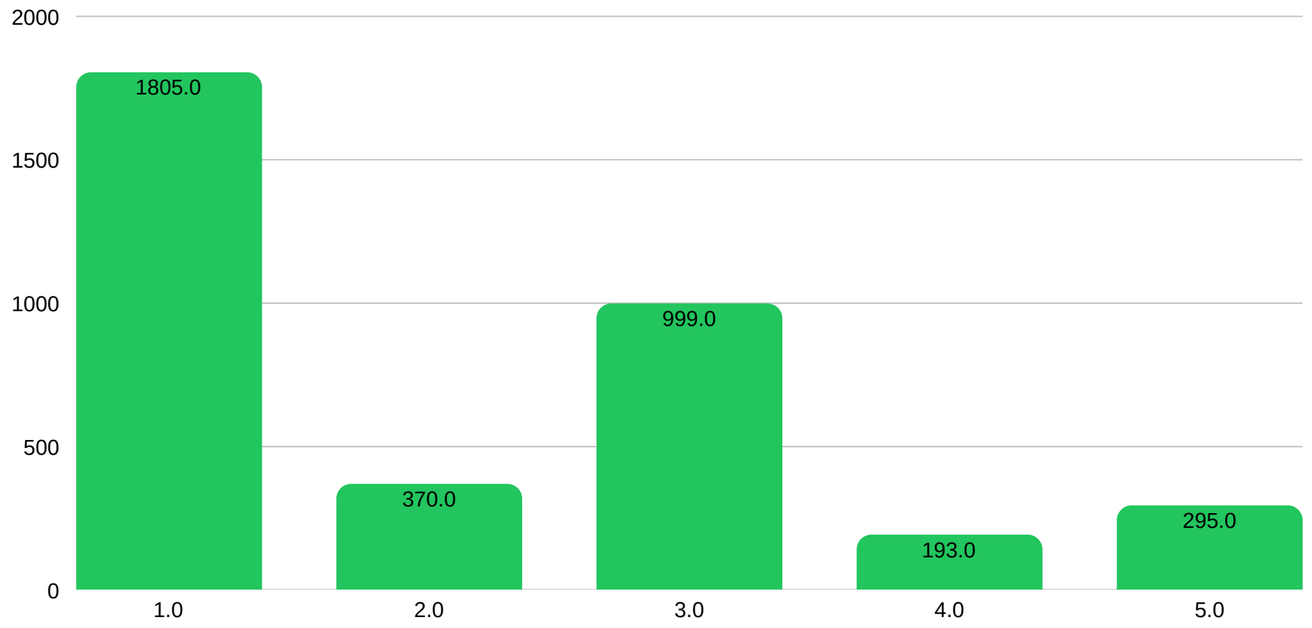


Photo and numbers are scored separately, then combined at the end (decision-level fusion).

DATA

Datasets we used

APTOS 2019 · 3,662 retina photos



Grade 0 is about half of all images → imbalance, handled during training.

IDRiD · Indian population, 3 parts

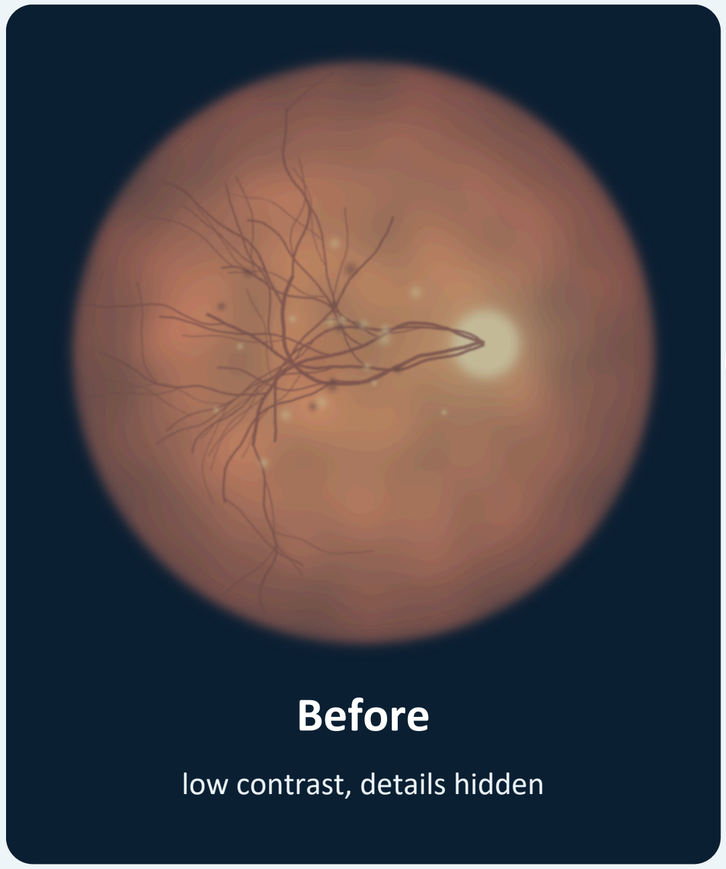
516 photos graded 0-4: our external test set

81 photos with lesion masks (54 train / 27 test)

516 photos with optic disc and fovea points

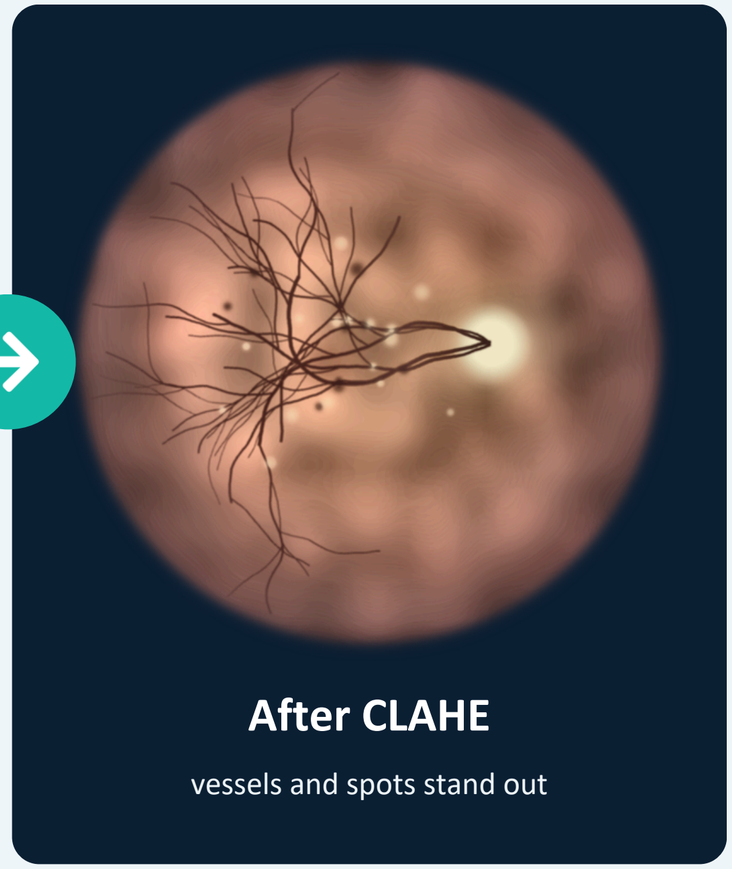
PREPROCESSING

Preparing the images



Before

low contrast, details hidden



After CLAHE

vessels and spots stand out

Illustration on a synthetic retina image - the same CLAHE operation is used on real photos.

1 Resize

224 px for grading, 512 px for lesion maps.

2 Boost contrast

CLAHE makes blood vessels and tiny lesions easier to see.

3 Normalize

Scale pixel values the way the pre-trained network expects.

Our classifier: how we trained it

1

Start smart

EfficientNet-B0, already pre-trained on millions of everyday images (transfer learning).

2

Fix imbalance

Show rare grades more often so the model learns them too.

3

Add variety

Random flips, rotation and brightness changes so it does not memorize.

4

Keep the best

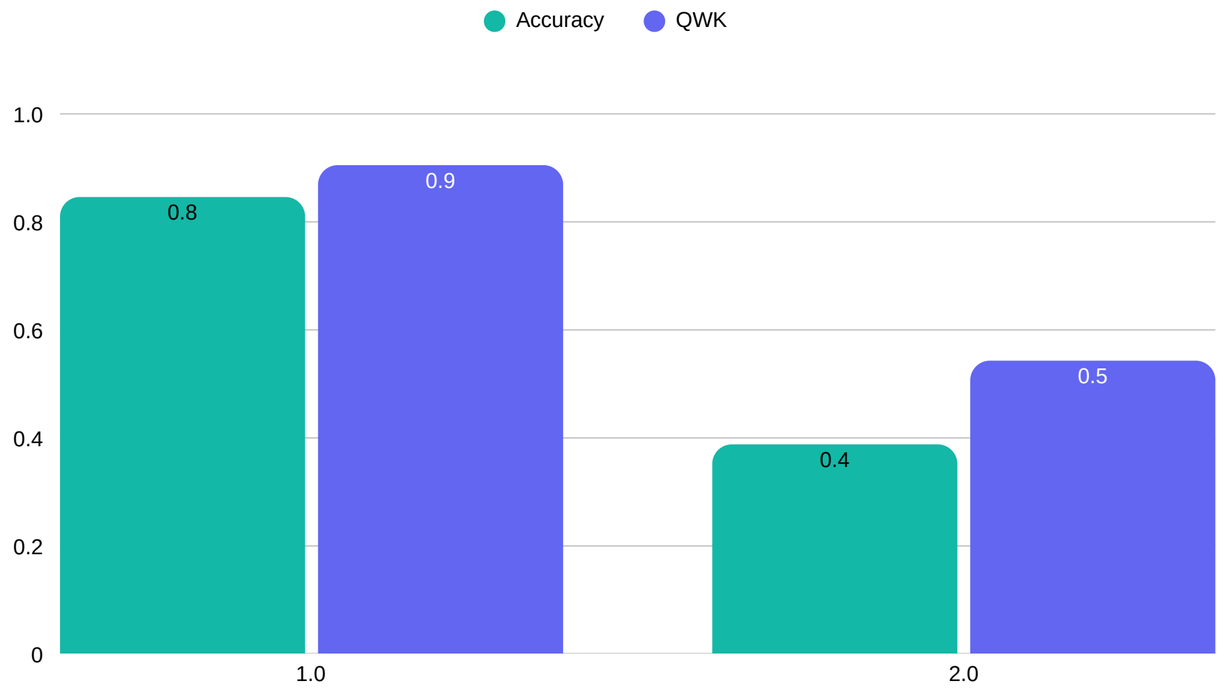
Train, check on held-out images, save the best epoch by QWK.

What is QWK?

A score for how well our grades agree with the doctors' grades. 1.0 is perfect, and a big mistake (0 vs 4) costs much more than a small one (1 vs 2).

EVALUATION

Results: strong, with an honest catch



APTOS · our test split

Strong

84.6%

accuracy

0.905

QWK

IDRiD · external test

Domain shift

38.8%

accuracy

0.543

QWK

Finding: very good on APTOS, a big drop on IDRiD (different camera + population). Weakest grades: Severe and Proliferative.

Explaining and locating

Grad-CAM heatmap

Warm colours show the areas that pushed the AI towards its grade. This builds trust.

Lesion maps (U-Net)

Colours mark the small signs of disease: microaneurysms, haemorrhages, hard and soft exudates.

Landmarks (ResNet)

Finds the optic disc and the fovea. Lesions near the fovea threaten sight the most.

Colour key used in the app

- Microaneurysms
- Haemorrhages
- Hard exudates
- Soft exudates
- Optic disc
- Fovea

Metabolic risk and the final decision

HbA1c

average blood sugar

Systolic BP

blood pressure

Duration

years with diabetes

0-3 pts each

0 - 9

total metabolic score, added from the three inputs

Low

score 0-2

Elevated

score 3-4

High

score 5-6

Urgent

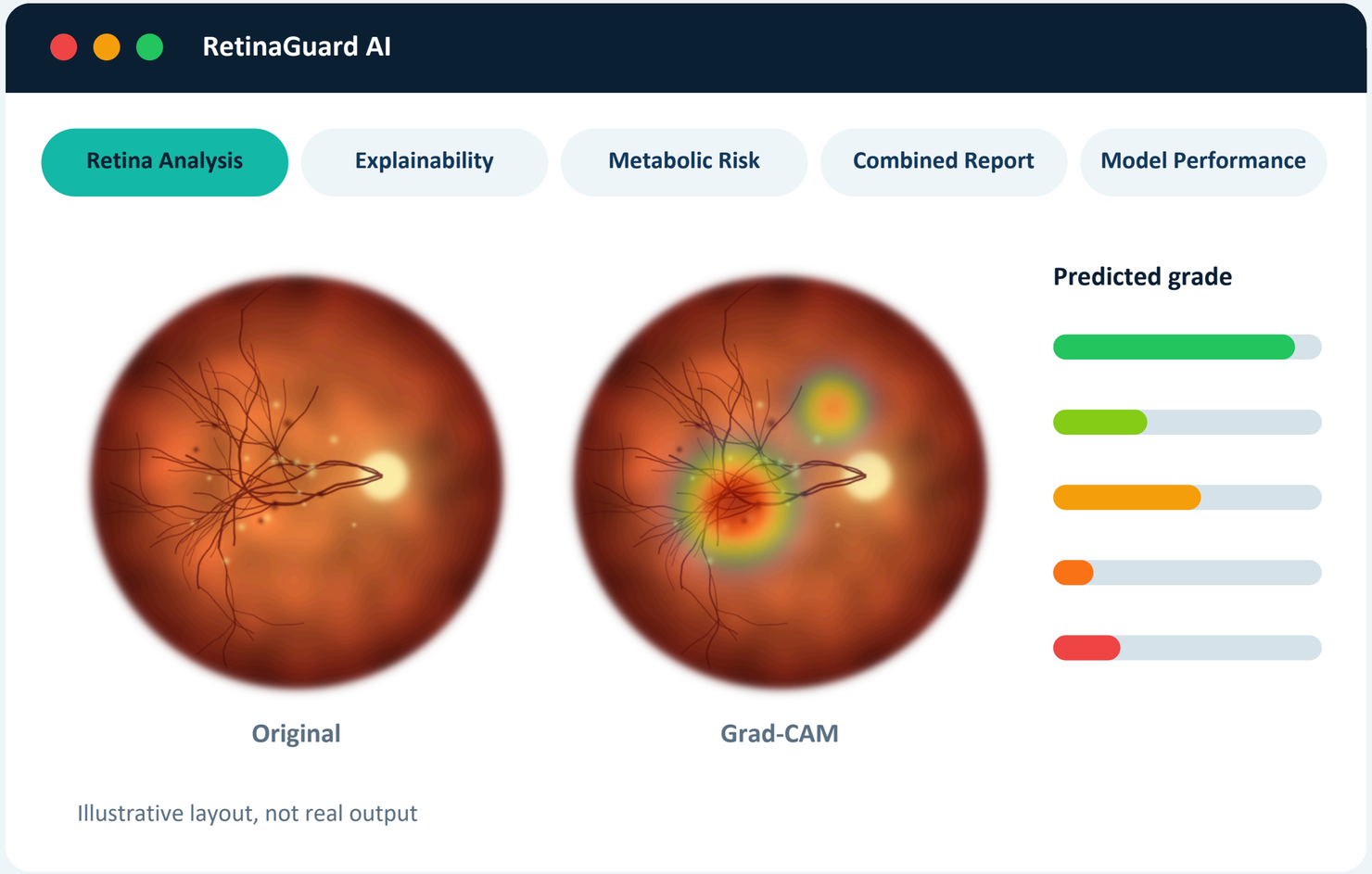
score 7-9

Final decision = 65% retina grade + 35% metabolic level

Severe (grade 3) is at least High. Proliferative (grade 4) is always Urgent.

LIVE DEMO COMES NEXT

The app



- Upload a photo or pick a sample
- Adjust heatmap and lesion opacity
- Looks good in light and dark mode
- Model status panel in the sidebar
- One-click PDF report

WHO DID WHAT

Team and contributions



Srushti

Person A · Data and vision models

- Data pipeline: APTOS + IDRiD
- Preprocessing (CLAHE)
- Lesion and landmark models



Prajakta

Person B · Classifier

- Trained EfficientNet-B0
- Tested on APTOS and IDRiD
- Grad-CAM and evaluation plots



Sanobar

Person C · App and risk

- Built the Streamlit app
- Risk score and fusion rules
- PDF report and deployment

HONESTY

Limitations: we are upfront

Separate data sources

Retina photos and blood values come from different datasets, so they are combined by rules, not learned together.

Domain shift

Accuracy drops from 84.6% on APTOS to 38.8% on IDRiD. Models need local validation.

Small data for lesions

Lesion maps learn from only 54 images, and tiny microaneurysms are often missed.

Not a medical device

This is a research prototype for learning. It must not be used for diagnosis.

ROADMAP

Next steps before the final review

1

Close the IDRiD gap

Fine-tune on mixed APTOS + IDRiD data and stronger augmentation.

2

Boost severe cases

Better handling of Severe and Proliferative with more targeted training.

3

Measure every model

Report Dice for lesion maps and pixel error for landmarks.

4

Go live and finish

Deploy the app online, test with users, write the final report.



Goal for the final review

A reliable model that works beyond one dataset, inside a live app, with full evaluation.

LIVE DEMO · QUESTIONS

Thank you

Let us show you DR Screening Suite live

Srushti - 21, Sanobar - 39, Prajakta -29

Research prototype for education. Not a medical device.

